

ERAY KULKIZAGA

Software Engineer | Python Backend | Applied AI

+90 539 603 39 41 | Istanbul, Türkiye

eraykulkizaga@hotmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

EXPERIENCE

FlyRank.ai | Back-End AI Engineering Intern (Remote)

Jul 2026 - Sep 2026

Usage Metering & Billing Engine | [GitHub](#)

- Built a PostgreSQL-first multi-tenant usage metering and billing service with FastAPI, SQLAlchemy, Alembic, monthly quotas, exact integer micro-USD pricing, durable alert jobs, and signature-verified Stripe test-mode subscription synchronization.
- Hardened retry, concurrency, and webhook correctness with idempotency keys, PostgreSQL advisory locks, payload fingerprinting, event-order guards, and tenant isolation. Verified that 12 concurrent usage retries produced exactly 1 persisted event and 8 concurrent webhook deliveries produced exactly 1 applied event across a 22-test PostgreSQL suite.

Microsoft Türkiye | AI Innovators Summer Internship (Remote)

Jun 2026 - Jul 2026

QueryPilot - Evidence-First PostgreSQL Performance Assistant | [GitHub](#) | [Live Demo](#)

- Built and released an evidence-first PostgreSQL performance assistant combining deterministic EXPLAIN-plan diagnostics, workload regression detection, and grounded RAG/LLM explanations.
- Implemented AST-based read-only SQL validation, pg_stat_statements workload ranking, plan baselines, and strict evidence and citation validation. Release evaluation achieved 12/12 rule-diagnosis, 12/12 no-answer behavior, and 9/9 Retrieval Hit@3, with 103 Python tests and 88.68% coverage.

ISKUR Holding | Enterprise Data Software Engineering Intern

Jun 2024 - Aug 2024

- Optimized SQL-based enterprise data retrieval using SAP Open SQL joins, reducing processing time by 41%.
- Developed backend validation and automation logic for contract, barcode, inventory, and operational-cost workflows, with transactional logging and structured audit trails.

SELECTED PROJECTS

Yazaris | Applied AI & Backend Engineer

Aug 2026 - Present

Selected for Stage 1 of TÜBİTAK 1812 BIGG

- Building an AI-assisted B2B order-intelligence system that converts unstructured Turkish orders into validated drafts using structured extraction, catalog matching, business rules, confidence gating, and human-in-the-loop review.
- Built a reproducible evaluation pipeline around a 640-SKU adversarial catalog and 60,000 synthetic orders. On a leak-free v0.4 benchmark of 6,066 programmatic cases, accepted drafts achieved 100% exact precision with 0 dangerous drafts; rejected a standalone structured-output LLM after it produced 3 dangerous drafts at 84% precision.

SupportFlow AI | Backend & Cloud Engineering - Team Project

Jul 2026 - Present

- Co-developed a multi-tenant FastAPI/PostgreSQL platform with database-verified RBAC, tenant-scoped ticketing, transactional audit trails, secure document ingestion, and retry-safe Redis/Celery processing; validated duplicate delivery and worker SIGKILL recovery across 200+ automated tests without duplicate side effects.
- Led and merged a protected GitHub Actions deployment workflow for AWS ECS using OIDC and immutable ECR image digests, with migration-before-promotion, atomic bootstrap behavior, upgrade and rollback guards, sanitized release evidence, and 16 contract and workflow tests.

SKILLS

Languages: Python, SQL

Backend: FastAPI, Pydantic, SQLAlchemy, Alembic, REST APIs, Celery, Webhooks, RBAC

Data: PostgreSQL, Redis

Applied AI: LLM APIs, RAG, Structured Outputs, LLM Evaluation, Human-in-the-Loop, MCP

Cloud & DevOps: AWS (ECS/Fargate, RDS, S3, ECR, CloudWatch), Docker, GitHub Actions, OIDC, Git

Testing & Quality: pytest, Integration Testing, Contract Testing, CI/CD

EDUCATION

Eastern Mediterranean University (EMU) - Famagusta, Northern Cyprus

B.Sc. in Software Engineering | Sep 2021 - Jun 2026

- ABET-accredited Software Engineering program | QS World University Rankings 2027: #=691
- English-medium instruction | Dean's High Honor & Honor List (2025-2026)

CERTIFICATIONS

DeepLearning.AI & Stanford Online - Machine Learning Specialization

Anthropic - Model Context Protocol: Advanced Topics

Anthropic - Building with the Claude API

LANGUAGES

English - C2 (CEFR), [EF SET 4-Skill Certificate](#) | **Turkish** - Native